Quantum Walk Search as a Model of Cognitive Search

From Marked Vertices to Quantum Cognition

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Abstract

This project bridges quantum walk search and quantum cognition. Its technical core is the Ambainis–Gilyén–Jeffery–Kokainis (AGJK) algorithm for finding marked vertices on an arbitrary graph, which achieves a quadratic speedup over the classical hitting time up to logarithmic factors. We interpret marked vertices as target memories or decision alternatives on a semantic or decision graph, and compare classical random walks, Szegedy/AGJK quantum walks, and open-system walks with partial measurement. We study how graph topology, marked-set structure, and decoherence affect hitting time, success probability, and choice probability, aiming to reproduce the $\widetilde{O}(\sqrt{\mathrm{HT}})$ scaling, relate the extended hitting time HT^+ to multi-alternative decision difficulty, and critically assess quantum cognition models.

1 Motivation and Scope

Quantum cognition models judgment and decision making with quantum probability, without claiming that the brain is a quantum computer; it captures contextuality, order effects, interference, and evidence accumulation that classical probability handles poorly [5, 6]. Quantum walks provide a process-level model: a walker on a semantic or decision graph represents unfolding evidence, and a marked vertex is a target memory or option [2, 3, 4].

The missing link is algorithmic structure. AGJK [1] does not merely detect a marked vertex; it finds one on any graph, for any marked set, with a quadratic speedup over the classical hitting time up to logarithmic factors. It also isolates the extended hitting time HT^+, which can be much larger than HT for multiple marked vertices. This maps naturally onto cognitive search with multiple targets and multi-alternative choice.

The project is anchored in the course's quantum-walk and query-complexity material; quantum cognition is the application, not a replacement for the QIS core. We do not claim the brain implements a quantum walk; we use quantum probability as a process-level framework and study its predictions and limits.

2 Research Questions and Hypotheses

RQ1 (mapping). How can marked-vertex search be mapped onto cognitive search on a semantic/decision graph, and what do HT, HT^+, spectral gap, and success probability correspond to cognitively (response time, choice probability, deliberation speed)?

RQ2 (structure). How do the number and arrangement of marked vertices (single, independent, clustered) change quantum versus classical search time and success probability? Can HT^+ formalize multi-alternative decision difficulty and context effects?

RQ3 (noise). How does partial measurement or decoherence erode the quantum advantage, and is there an optimal "introspection" level that maximizes success probability or decision speed?

We will test three hypotheses. **H1:** for well-connected graphs and a single target, quantum search time scales as $\sqrt{\mathrm{HT}}$ and beats the classical baseline. **H2:** clustered multiple targets can cause a large HT^+ penalty for a naive quantum search, while the AGJK improved algorithm restores $\widetilde{O}(\sqrt{\mathrm{HT}})$ scaling. **H3:** choice probability and response time depend non-monotonically on target similarity or clustering, analogous to similarity and attraction effects; weak partial measurement may help by suppressing destructive interference, while strong measurement destroys the advantage.

3 Approach and Methods

3.1 Quantum information core

We will review and reproduce Szegedy's walk and AGJK: the discriminant matrix $D_{xy} = \sqrt{P_{xy}P^*_{yx}}$, the walk operator, HT, HT^+, the interpolated walk $P(s) = (1-s)P + sP'$, and quantum fast-forwarding [1]. Numerically, we will implement Algorithm 1 and test Conjecture 3 on small graphs; Theorem 5 will be cited as the guarantee, and its logarithmic overhead discussed rather than re-proved.

3.2 Cognitive mapping

A graph $G = (V, E)$ represents concepts, evidence states, or options; a marked set $M \subset V$ is a target memory or decision boundary. From an initial state $|\psi_0\rangle$, the walk evolves unitarily; success probability is $p_M(t) = \|\Pi_M U(t)|\psi_0\rangle\|^2$, response time is the first t with $p_M(t) \geq p^*$, and for multiple marked sets $M_1, \ldots, M_k$ the choice probability for option i is the mass in M_i at decision time. The classical baseline is a Markov walk on the same graph. This is a process-level model, not a biological claim.

3.3 Experiments

We will use synthetic graphs (complete, cycle, 2D grid, hypercube, random regular, clustered) and a small hand-built semantic graph; if time permits, a ConceptNet subgraph following Pavone–Faro [4]. Marked-set configurations: single, independent, clustered, and absorbing decision boundaries. Metrics: classical/quantum search time, success probability vs. time, scaling exponents, spectral gap, choice probability, and response time. A partial-measurement model will interpolate between unitary and classical behavior. The implementation will use Python, NumPy/SciPy, NetworkX, and Matplotlib, with fixed seeds, configs, and a README reproducing every figure.

4 Expected Results and Contributions

We expect (i) numerical reproduction of $\widetilde{O}(\sqrt{\mathrm{HT}})$ scaling on at least two graph families; (ii) a comparison of single, independent, and clustered marked sets, showing when the HT^+ penalty appears and how the improved algorithm avoids it; (iii) a semantic-graph demonstration where quantum and classical walks produce different target-activation and response-time patterns; and (iv) a decoherence sweep showing whether a nonzero measurement rate helps or hurts. The contribution is not a new theorem but a careful bridge: a rigorous quantum algorithm becomes a testable process model for cognitive search and multi-alternative choice. The discussion will separate algorithmic advantage from biological plausibility and state what would be needed to fit behavioral data.

5 Timeline

Dates	Milestones
Oct 21	Proposal submitted; scope confirmed with instructor.
Oct 22–Nov 4	Read AGJK and Chen et al.; implement classical and Szegedy walks; reproduce basic scaling.
Nov 5–Nov 18	Build semantic and decision graphs; implement cognitive mapping; run single vs. multiple/clustered marked-set experiments.
Nov 19–Nov 25	Add partial measurement/decoherence; complete sensitivity analysis.
Nov 26–Dec 9	Freeze figures; write theory, results, and discussion; complete a full draft.
Dec 10–Dec 13	Prepare and rehearse slides.
Dec 14	Project presentation.
Dec 15–Dec 21	Incorporate feedback; submit the final paper.

References

[1] A. Ambainis, A. Gilyén, S. Jeffery, and M. Kokainis, "Quadratic speedup for finding marked vertices by quantum walks," arXiv:1903.07493, 2019.

[2] M. Chen, G. M. Ferro, and D. Sornette, "On the use of discrete-time quantum walks in decision theory," *PLoS ONE* 17(8), e0273551, 2022.

[3] J. Huang and J. R. Busemeyer, "A quantum walk framework for multialternative decision making," ICCM 2022; DOI: 10.31234/osf.io/hmv4a.

[4] A. Pavone and S. Faro, "Modeling the emergence of insight via quantum interference on semantic graphs," *Mathematics* 13(19), 3171, 2025.

[5] E. M. Pothos and J. R. Busemeyer, "Quantum cognition," *Annual Review of Psychology* 73, 749–778, 2022.

[6] J. R. Busemeyer and P. D. Bruza, *Quantum Models of Cognition and Decision*, 2nd ed., Cambridge University Press, 2024.